

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.4	7.1	0.3	0	0	66.2
U3 - Toxicity prior Beta(0.15, 0.85)	26.9	5.2	0.5	0	0	67.4
U2 - Toxicity prior Beta(0.3, 0.7)	27.2	7.5	0.4	0	0	64.9
U3 - Toxicity prior Beta(0.3, 0.7)	27.4	6.6	0.4	0	0	65.6
U2 - Toxicity prior Beta(0.5, 0.5)	30.1	9.5	0.7	0	0	59.7
U3 - Toxicity prior Beta(0.5, 0.5)	30.1	9	0.3	0	0	60.6
U2 - Toxicity prior Beta(1, 1)	29.6	9.5	0.7	0	0	60.2
U3 - Toxicity prior Beta(1, 1)	29.8	9.3	0.6	0	0	60.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.9	5.1	1	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	16.2	4.8	1	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	16	5.3	1.1	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	16.2	5	1.1	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	15.7	5.7	1.6	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	15.9	5.4	1.6	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	15.7	5.7	1.7	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	15.9	5.5	1.7	0.6	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3	0.4	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1	2.8	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3	0.4	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1	2.9	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3	0.5	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	2.9	0.5	0.3	0	
U2 - Toxicity prior Beta(1, 1)	1.1	3	0.5	0.3	0	
U3 - Toxicity prior Beta(1, 1)	1.1	2.9	0.5	0.3	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.4	29.3	31.9	7.1	0.3	1
U3 - Toxicity prior Beta(0.15, 0.85)	33	30	30.7	5.3	0.1	0.9
U2 - Toxicity prior Beta(0.3, 0.7)	27.8	29.4	33.3	8.2	0.3	1
U3 - Toxicity prior Beta(0.3, 0.7)	30.8	29.7	31.5	6.9	0.2	0.9
U2 - Toxicity prior Beta(0.5, 0.5)	24.1	25.9	38.8	9.8	0.4	1
U3 - Toxicity prior Beta(0.5, 0.5)	28.1	27.3	34.8	8.4	0.3	1.1
U2 - Toxicity prior Beta(1, 1)	24	25.8	38.2	10.5	0.5	1
U3 - Toxicity prior Beta(1, 1)	27.6	27.5	35.8	7.7	0.4	1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11	9.2	7.5	2	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	11.5	9.4	7.1	1.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	10.7	9.1	7.7	2.2	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	11.2	9.2	7.4	1.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	9.3	9.3	8.6	2.5	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	9.8	9.5	8.2	2.2	0.1	
U2 - Toxicity prior Beta(1, 1)	9.2	9.4	8.7	2.5	0.2	
U3 - Toxicity prior Beta(1, 1)	9.7	9.6	8.3	2.2	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.8	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.9	1.6	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.9	1.8	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.8	1.7	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	2	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.6	1.9	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.5	2	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.8	3.5	1.9	0.5	0.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13	54.7	23.4	5.2	0.1	3.6
U3 - Toxicity prior Beta(0.15, 0.85)	14.1	57.9	20	4.4	0	3.6
U2 - Toxicity prior Beta(0.3, 0.7)	11.5	54	25.6	5.2	0.3	3.4
U3 - Toxicity prior Beta(0.3, 0.7)	13.4	58	20.5	4.5	0.1	3.5
U2 - Toxicity prior Beta(0.5, 0.5)	10.2	51.5	27.8	6.8	0.6	3.1
U3 - Toxicity prior Beta(0.5, 0.5)	11	57.7	22.7	5.4	0.1	3.1
U2 - Toxicity prior Beta(1, 1)	10.4	51.7	27.6	6.8	0.5	3
U3 - Toxicity prior Beta(1, 1)	11.7	56.1	23.1	5.8	0.3	3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.8	12.7	6	1.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.1	13.3	5.4	1.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.3	12.9	6.2	2.1	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8.7	13.3	5.7	1.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	7.1	13	7	2.4	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	7.3	13.8	6.4	2.1	0.2	
U2 - Toxicity prior Beta(1, 1)	7	13.1	7.1	2.3	0.2	
U3 - Toxicity prior Beta(1, 1)	7.2	13.7	6.5	2.1	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	4.5	1.7	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	4.5	1.6	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	4.5	1.8	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	4.5	1.5	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.9	1.9	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	4	1.7	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	3.9	1.9	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	4	1.7	0.6	0.1	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.3	8.7	25	49.8	7.9	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	10.9	11.7	21.8	48.6	6.8	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	6.1	7.8	23.6	52.9	9.4	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	11.3	20.1	51.6	7.7	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	5.7	7.7	18.5	57.2	10.7	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	8.6	9.6	17.9	54.5	9.2	0.2
U2 - Toxicity prior Beta(1, 1)	4.6	8.2	19.2	56.6	11.2	0.2
U3 - Toxicity prior Beta(1, 1)	7.9	10.6	18	54.2	9.1	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	5.5	7.8	10	2.1	
U3 - Toxicity prior Beta(0.15, 0.85)	5	5.5	7.7	10	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	5.1	7.7	10.5	2.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.7	5.3	7.5	10.4	2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	4.9	7.2	11.3	2.6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.3	5	7.2	11.2	2.2	
U2 - Toxicity prior Beta(1, 1)	4	4.9	7.2	11.2	2.7	
U3 - Toxicity prior Beta(1, 1)	4.3	5.1	7.3	11.1	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.2	1.8	2.5	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.2	1.8	2.5	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.1	1.8	2.5	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3.1	1.8	2.3	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.7	2.4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.7	2.3	0.6	
U2 - Toxicity prior Beta(1, 1)	0.2	2.9	1.7	2.4	0.7	
U3 - Toxicity prior Beta(1, 1)	0.3	3	1.7	2.3	0.6	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	34.6	27.5	24.5	8.5	0.5
U3 - Toxicity prior Beta(0.15, 0.85)	6.6	43.3	25.8	18.5	5.3	0.5
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	33.3	27.6	25.2	9	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	6.5	41.1	28.2	18.6	5.2	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	32.5	27.9	24.8	10.1	0.4
U3 - Toxicity prior Beta(0.5, 0.5)	6.2	39.9	28	19.1	6.4	0.4
U2 - Toxicity prior Beta(1, 1)	3.9	31.1	29.9	24.1	10.6	0.4
U3 - Toxicity prior Beta(1, 1)	6.5	38.2	29.1	18.8	6.9	0.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.3	8.4	7.5	6.9	2.9	
U3 - Toxicity prior Beta(0.15, 0.85)	4.5	9.1	7.6	6.5	2.3	
U2 - Toxicity prior Beta(0.3, 0.7)	4.2	8.1	7.4	7	3.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4.4	8.8	7.8	6.6	2.4	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	7.8	7.5	7.4	3.5	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	8.6	7.9	6.8	2.7	
U2 - Toxicity prior Beta(1, 1)	3.8	7.7	7.6	7.4	3.6	
U3 - Toxicity prior Beta(1, 1)	3.9	8.4	7.8	7	2.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.5	2.1	2.1	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.5	2.1	2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.4	2.1	2.2	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.4	2.2	2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3.1	2.1	2.2	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	2.2	2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.1	3.1	2.1	2.2	1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.1	2.2	2	0.8	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	66.2	14.5	1.5	0.3	0	17.5
U3 - Toxicity prior Beta(0.15, 0.85)	69.4	12.4	0.5	0.3	0	17.4
U2 - Toxicity prior Beta(0.3, 0.7)	67	15.2	1.5	0.5	0	15.8
U3 - Toxicity prior Beta(0.3, 0.7)	69.1	13.7	0.7	0.5	0	16
U2 - Toxicity prior Beta(0.5, 0.5)	64.9	18.2	1.9	0.7	0	14.3
U3 - Toxicity prior Beta(0.5, 0.5)	67.6	16.2	1.1	0.6	0	14.5
U2 - Toxicity prior Beta(1, 1)	66.2	17.1	1.7	0.6	0	14.4
U3 - Toxicity prior Beta(1, 1)	68.5	15.7	1.2	0.4	0	14.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.8	5.9	1.1	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20.1	5.8	0.9	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	19.8	6	1.2	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.1	5.9	1	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	18.8	6.7	1.6	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	19.3	6.6	1.3	0.3	0	
U2 - Toxicity prior Beta(1, 1)	18.7	6.8	1.6	0.3	0	
U3 - Toxicity prior Beta(1, 1)	19.2	6.6	1.3	0.3	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.7	0.4	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.6	0.3	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.7	0.4	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.6	0.3	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.6	0.6	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	2.5	0.4	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.8	2.6	0.6	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.9	2.6	0.4	0.1	0	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.1	59.8	20	3.3	0.6	1.2
U3 - Toxicity prior Beta(0.15, 0.85)	16.4	62.5	16.9	2.6	0.4	1.2
U2 - Toxicity prior Beta(0.3, 0.7)	13.2	59.9	21.4	3.9	0.7	0.9
U3 - Toxicity prior Beta(0.3, 0.7)	15.8	63.6	16.2	2.8	0.5	1.1
U2 - Toxicity prior Beta(0.5, 0.5)	11.3	58.4	24	5	0.5	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	13.5	63.5	18	3.8	0.3	0.9
U2 - Toxicity prior Beta(1, 1)	11.8	57.6	23.9	5.3	0.5	0.9
U3 - Toxicity prior Beta(1, 1)	13.3	63.5	18.1	3.7	0.3	1.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.5	15	4.4	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	15.5	3.8	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	9.2	15	4.6	1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.5	15.6	3.8	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	8	15.5	5.3	1.1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.4	16.3	4.2	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	7.9	15.5	5.3	1.1	0.1	
U3 - Toxicity prior Beta(1, 1)	8.2	16.3	4.3	0.9	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	4.1	1.1	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	4.2	0.9	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	4.1	1.2	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	4.1	0.9	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.6	1.3	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.7	1	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.4	3.6	1.3	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.5	3.6	1	0.2	0	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	44.4	5.7	0.6	0.1	0.1	49.1
U3 - Toxicity prior Beta(0.15, 0.85)	44.9	5.5	0.5	0.1	0	49
U2 - Toxicity prior Beta(0.3, 0.7)	44.9	6.9	0.6	0.1	0.1	47.4
U3 - Toxicity prior Beta(0.3, 0.7)	45.8	7.3	0.3	0.1	0	46.5
U2 - Toxicity prior Beta(0.5, 0.5)	46.8	9.4	0.5	0.1	0.1	43.1
U3 - Toxicity prior Beta(0.5, 0.5)	46.7	8.9	0.3	0.1	0	44
U2 - Toxicity prior Beta(1, 1)	46.6	8.4	0.9	0.1	0.1	43.9
U3 - Toxicity prior Beta(1, 1)	47.1	9.1	0.3	0.1	0	43.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.5	4.9	0.8	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	17.7	4.7	0.8	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	17.7	5	0.9	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	17.8	4.9	0.9	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17.2	5.6	1.4	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	17.3	5.4	1.3	0.3	0	
U2 - Toxicity prior Beta(1, 1)	17.1	5.6	1.4	0.3	0	
U3 - Toxicity prior Beta(1, 1)	17.3	5.5	1.4	0.3	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	2.5	0.3	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.5	0.3	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	2.6	0.3	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.5	0.3	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1	2.7	0.4	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	2.6	0.4	0.1	0	
U2 - Toxicity prior Beta(1, 1)	1	2.6	0.4	0.1	0	
U3 - Toxicity prior Beta(1, 1)	1	2.6	0.4	0.1	0	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20	23	22.6	13.2	11.6	9.6
U3 - Toxicity prior Beta(0.15, 0.85)	24.6	24.6	19.9	11.9	9.6	9.4
U2 - Toxicity prior Beta(0.3, 0.7)	18.5	22.5	23.5	13.4	14.3	7.8
U3 - Toxicity prior Beta(0.3, 0.7)	24.8	23	20	12.7	11.5	8
U2 - Toxicity prior Beta(0.5, 0.5)	17.2	21.4	23.8	16.3	16.8	4.5
U3 - Toxicity prior Beta(0.5, 0.5)	24.9	23	20.1	13.3	14	4.7
U2 - Toxicity prior Beta(1, 1)	17.1	22.8	24.1	14.6	16.9	4.5
U3 - Toxicity prior Beta(1, 1)	23.9	23.3	20.4	13.9	13.6	4.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	7.5	5.8	4.3	2.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	7.3	5.6	4.3	2.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	7.4	6	4.5	2.7	
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	7.1	5.7	4.4	2.5	
U2 - Toxicity prior Beta(0.5, 0.5)	7.6	7.2	6.7	4.9	3.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	7.1	6.3	4.9	3	
U2 - Toxicity prior Beta(1, 1)	7.5	7.2	6.7	4.9	3.2	
U3 - Toxicity prior Beta(1, 1)	8.3	7.1	6.4	4.9	3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.2	1.7	1.5	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.1	1.6	1.5	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.1	1.7	1.5	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	4.1	1.6	1.5	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.8	1.8	1.6	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.8	1.7	1.6	0.7	
U2 - Toxicity prior Beta(1, 1)	0.7	3.7	1.8	1.6	0.7	
U3 - Toxicity prior Beta(1, 1)	0.8	3.7	1.7	1.6	0.7	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.3	34	9.3	1.4	0.1	33.9
U3 - Toxicity prior Beta(0.15, 0.85)	22.2	33.4	9	1	0.1	34.3
U2 - Toxicity prior Beta(0.3, 0.7)	20	36.4	9.9	1.9	0.2	31.6
U3 - Toxicity prior Beta(0.3, 0.7)	20	38.2	8.9	1.8	0.1	31
U2 - Toxicity prior Beta(0.5, 0.5)	18.8	40.4	13.8	2.2	0.2	24.6
U3 - Toxicity prior Beta(0.5, 0.5)	20.3	41.2	11.9	1.9	0.2	24.5
U2 - Toxicity prior Beta(1, 1)	19.4	40.4	12.9	2.6	0.2	24.5
U3 - Toxicity prior Beta(1, 1)	20.6	41	11.2	2.2	0.2	24.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	13	9.8	3.1	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	13.3	10	2.8	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	12.8	10.1	3.3	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	13	10.2	3	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	11.1	11	4.2	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	11.3	11.3	4	1	0.1	
U2 - Toxicity prior Beta(1, 1)	11.1	11	4.3	1.1	0.1	
U3 - Toxicity prior Beta(1, 1)	11.3	11.3	4	1	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.4	0.9	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	4.3	0.8	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	4.4	0.9	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	4.3	0.8	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.9	1.1	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.9	1.1	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.6	3.9	1.1	0.4	0	
U3 - Toxicity prior Beta(1, 1)	0.6	3.9	1	0.3	0	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	67	22.4	1.8	0.1	0	8.7
U3 - Toxicity prior Beta(0.15, 0.85)	70.2	20.5	1.6	0	0	7.7
U2 - Toxicity prior Beta(0.3, 0.7)	66.7	22.9	2.8	0.1	0	7.5
U3 - Toxicity prior Beta(0.3, 0.7)	69.3	21.4	1.7	0	0	7.6
U2 - Toxicity prior Beta(0.5, 0.5)	65.1	25.5	3.2	0.1	0	6.1
U3 - Toxicity prior Beta(0.5, 0.5)	66.2	25.4	2.1	0	0	6.3
U2 - Toxicity prior Beta(1, 1)	64.2	26.1	2.9	0.1	0	6.7
U3 - Toxicity prior Beta(1, 1)	68.1	23.4	2.1	0	0	6.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.7	7.8	1.8	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	19.2	7.6	1.7	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	18.4	8	2	0.6	0	
U3 - Toxicity prior Beta(0.3, 0.7)	18.8	7.8	1.8	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17.3	8.7	2.5	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17.6	8.5	2.3	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	17.2	8.7	2.6	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	17.7	8.4	2.4	0.6	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3.8	0.7	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.7	0.6	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.8	0.7	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.8	0.7	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	3.7	0.9	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	3.6	0.8	0.3	0	
U2 - Toxicity prior Beta(1, 1)	1.2	3.7	0.9	0.3	0	
U3 - Toxicity prior Beta(1, 1)	1.2	3.6	0.8	0.2	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.4	17.1	14.4	42.9	12.8	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	16.3	19.7	12.4	40.3	10.9	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	11.8	15.5	13.1	43.7	15.5	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	16.3	17.6	12.4	40.6	12.7	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	10.2	13.4	12.1	46.9	16.9	0.5
U3 - Toxicity prior Beta(0.5, 0.5)	14.9	17.4	9.9	43.1	14.4	0.3
U2 - Toxicity prior Beta(1, 1)	9.9	12.9	12.6	47	17.2	0.4
U3 - Toxicity prior Beta(1, 1)	15.6	16	10.6	43.2	14.1	0.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.1	6.3	6.3	8.8	3.4	
U3 - Toxicity prior Beta(0.15, 0.85)	5.7	6.4	6.1	8.7	3.1	
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	6	6.1	9.3	3.8	
U3 - Toxicity prior Beta(0.3, 0.7)	5.4	6.1	5.9	9.1	3.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	5.6	5.9	9.8	4.2	
U3 - Toxicity prior Beta(0.5, 0.5)	5	5.7	5.7	9.7	3.8	
U2 - Toxicity prior Beta(1, 1)	4.5	5.7	5.9	9.7	4.2	
U3 - Toxicity prior Beta(1, 1)	5	5.7	5.8	9.7	3.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.4	1.7	2.3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.5	1.7	2.2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.4	1.7	2.3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.4	1.7	2.2	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.2	1.7	2.2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.7	2.2	0.9	
U2 - Toxicity prior Beta(1, 1)	0.3	3.2	1.7	2.2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.5	3.2	1.7	2.2	0.9	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.7	36.1	8.6	2.4	0.2	30
U3 - Toxicity prior Beta(0.15, 0.85)	26.1	35.1	6.8	2	0.2	29.8
U2 - Toxicity prior Beta(0.3, 0.7)	21.8	36.8	10	2.6	0.4	28.4
U3 - Toxicity prior Beta(0.3, 0.7)	23.9	37.5	8.3	2.4	0.2	27.7
U2 - Toxicity prior Beta(0.5, 0.5)	20	41.2	12.8	3.3	0.6	22.1
U3 - Toxicity prior Beta(0.5, 0.5)	21.7	41.1	11.7	2.6	0.6	22.3
U2 - Toxicity prior Beta(1, 1)	20	40.6	13	3.6	0.5	22.3
U3 - Toxicity prior Beta(1, 1)	22.6	40.8	10.9	2.8	0.5	22.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.9	9.4	3.4	1.3	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	13.2	9.5	3.2	1.2	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	12.5	9.6	3.7	1.4	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	12.7	9.8	3.4	1.3	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	11	10.4	4.6	1.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	11.2	10.6	4.3	1.6	0.2	
U2 - Toxicity prior Beta(1, 1)	10.9	10.4	4.7	1.7	0.2	
U3 - Toxicity prior Beta(1, 1)	11.2	10.6	4.4	1.6	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.6	0.9	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.5	0.9	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.6	1	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	4.6	1	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	4.2	1.2	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	4.1	1.1	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	4.1	1.2	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.8	4.1	1.1	0.6	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23.1	22.7	21.1	23.1	8.3	1.7
U3 - Toxicity prior Beta(0.15, 0.85)	30.8	26.5	14.2	20.6	6.4	1.5
U2 - Toxicity prior Beta(0.3, 0.7)	19.7	21.8	20.8	27.2	9	1.5
U3 - Toxicity prior Beta(0.3, 0.7)	26.8	25.5	15.7	23.9	6.9	1.2
U2 - Toxicity prior Beta(0.5, 0.5)	16.7	21.1	20	30.7	10.6	0.9
U3 - Toxicity prior Beta(0.5, 0.5)	24.5	25.7	12.8	27.8	8.2	1
U2 - Toxicity prior Beta(1, 1)	17.5	21.6	19.5	29.8	10.8	0.8
U3 - Toxicity prior Beta(1, 1)	26	25.5	12.8	26.5	8.4	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.1	7.4	6.4	6.5	2.4	
U3 - Toxicity prior Beta(0.15, 0.85)	7.8	7.4	6.1	6.3	2.3	
U2 - Toxicity prior Beta(0.3, 0.7)	6.7	7.1	6.4	6.9	2.8	
U3 - Toxicity prior Beta(0.3, 0.7)	7.5	7.1	6	6.7	2.6	
U2 - Toxicity prior Beta(0.5, 0.5)	6.1	6.8	6.3	7.4	3.3	
U3 - Toxicity prior Beta(0.5, 0.5)	6.8	6.8	6	7.3	3	
U2 - Toxicity prior Beta(1, 1)	5.9	7	6.4	7.4	3.2	
U3 - Toxicity prior Beta(1, 1)	6.8	6.8	6	7.2	3.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4	1.8	2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.8	1.9	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.9	1.9	2.1	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.9	1.8	2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.6	1.8	2.1	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.7	1.7	2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.6	3.6	1.8	2.1	0.8	
U3 - Toxicity prior Beta(1, 1)	0.8	3.6	1.7	2	0.8	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	47.7	36.7	10.2	2.1	0.1	3.2
U3 - Toxicity prior Beta(0.15, 0.85)	52.8	34.3	7.6	1.5	0	3.8
U2 - Toxicity prior Beta(0.3, 0.7)	45.7	36.5	12.3	2.8	0.2	2.5
U3 - Toxicity prior Beta(0.3, 0.7)	53	34.2	8.3	1.6	0	2.9
U2 - Toxicity prior Beta(0.5, 0.5)	45	37	11	4	0.5	2.5
U3 - Toxicity prior Beta(0.5, 0.5)	53	34.8	8.2	1.5	0.2	2.3
U2 - Toxicity prior Beta(1, 1)	46.4	36.9	10.6	3.4	0.4	2.3
U3 - Toxicity prior Beta(1, 1)	53.5	35.3	7.4	1.4	0.1	2.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.8	9.6	5	2.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	12.7	9.2	4.6	2.6	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	11.3	9.4	5.2	3	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	12.3	9	4.8	2.9	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	10.4	9.6	5.6	3.2	1	
U3 - Toxicity prior Beta(0.5, 0.5)	11.4	9.2	5.3	3.1	0.9	
U2 - Toxicity prior Beta(1, 1)	10.4	9.6	5.7	3.2	1	
U3 - Toxicity prior Beta(1, 1)	11.3	9.3	5.3	3	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	4.5	1.7	1.1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	1.3	4.4	1.6	1.1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	4.5	1.8	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	1.3	4.3	1.7	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	4.3	1.8	1.3	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	1.4	4.2	1.7	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	1.2	4.3	1.8	1.3	0.3	
U3 - Toxicity prior Beta(1, 1)	1.4	4.2	1.7	1.2	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	31.5	30.6	16.5	13.6	7.8	0
U3 - Toxicity prior Beta(0.15, 0.85)	37.1	32.4	14.1	10.8	5.6	0
U2 - Toxicity prior Beta(0.3, 0.7)	32.5	31	15.5	13.2	7.8	0
U3 - Toxicity prior Beta(0.3, 0.7)	36.8	32	15.2	10.2	5.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	33	30.3	16.7	12.7	7.3	0
U3 - Toxicity prior Beta(0.5, 0.5)	37	31.3	15.2	10.7	5.8	0
U2 - Toxicity prior Beta(1, 1)	33.6	29.8	17.4	11.6	7.6	0
U3 - Toxicity prior Beta(1, 1)	37.1	31.4	15.9	9.9	5.7	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.2	5.6	6.3	8.2	5.7	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	6.2	6.4	7.8	5.1	
U2 - Toxicity prior Beta(0.3, 0.7)	4.2	5.6	6.2	8.1	5.9	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	6.1	6.3	7.8	5.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	5.4	6.2	8.3	6.1	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	6	6.3	8	5.3	
U2 - Toxicity prior Beta(1, 1)	4.1	5.4	6.2	8.3	6.1	
U3 - Toxicity prior Beta(1, 1)	4.4	5.9	6.4	7.9	5.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	2.3	2.9	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2.4	2.7	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	2.3	2.9	1.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	2.4	2.7	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	2.3	2.9	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	2.4	2.8	1.5	
U2 - Toxicity prior Beta(1, 1)	0.1	2.7	2.3	2.9	1.6	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	2.4	2.8	1.4	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18	69.3	12.4	0.2	0	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	22.3	68.7	8.7	0.2	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	17.5	68.2	13.7	0.3	0	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	19.8	69.2	10.7	0.2	0	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	15.6	68	16.1	0.2	0	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	19.2	69.3	11.1	0.3	0	0.1
U2 - Toxicity prior Beta(1, 1)	16.5	67.8	15.4	0.2	0	0.1
U3 - Toxicity prior Beta(1, 1)	20.6	67.1	11.9	0.3	0	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.2	16.3	3.3	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	10.8	16.5	2.5	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	9.8	16.3	3.5	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	10.6	16.5	2.7	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	16.8	3.9	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	9.6	17.2	2.9	0.2	0	
U2 - Toxicity prior Beta(1, 1)	9	16.9	3.8	0.3	0	
U3 - Toxicity prior Beta(1, 1)	9.7	17.1	2.9	0.2	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.2	1	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4.2	0.7	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	4.2	1.1	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	4.2	0.7	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.8	1.1	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.9	0.7	0.1	0	
U2 - Toxicity prior Beta(1, 1)	0.6	3.8	1.1	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.7	3.9	0.8	0.1	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.4	11.7	9.6	21.7	47.6	0
U3 - Toxicity prior Beta(0.15, 0.85)	14.9	15.3	10.1	18.6	41.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.9	11.1	10.2	19.5	49.3	0
U3 - Toxicity prior Beta(0.3, 0.7)	15	13.8	10.5	17.2	43.5	0
U2 - Toxicity prior Beta(0.5, 0.5)	8.6	10	9.2	18.8	53.4	0
U3 - Toxicity prior Beta(0.5, 0.5)	14.6	12.3	11	15.9	46.2	0
U2 - Toxicity prior Beta(1, 1)	8.8	8.8	9.8	18	54.6	0
U3 - Toxicity prior Beta(1, 1)	14.2	12.8	10.8	16.3	45.9	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.4	4.2	5.3	7.6	9.4	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.4	5.5	7.7	8.8	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	4.1	5.1	7.6	9.8	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	4.2	5.3	7.7	9.3	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.9	4.9	7.6	10.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	4	5.2	7.7	9.6	
U2 - Toxicity prior Beta(1, 1)	3.3	3.9	5	7.5	10.4	
U3 - Toxicity prior Beta(1, 1)	3.4	4	5.2	7.8	9.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.5	2.5	1.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.6	2.6	1.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.5	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.5	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.5	1.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.5	1.7	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.5	1.8	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.6	2.5	1.7	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.6	60	12.6	0.4	0	4.4
U3 - Toxicity prior Beta(0.15, 0.85)	24.6	59.6	10.2	0.3	0	5.3
U2 - Toxicity prior Beta(0.3, 0.7)	18.9	61.2	15.6	0.4	0	3.9
U3 - Toxicity prior Beta(0.3, 0.7)	20.4	62.4	12.4	0.5	0	4.3
U2 - Toxicity prior Beta(0.5, 0.5)	17.7	61.4	17.4	0.5	0	3
U3 - Toxicity prior Beta(0.5, 0.5)	18.7	63.4	14.4	0.5	0	3
U2 - Toxicity prior Beta(1, 1)	17	61.9	17.2	0.5	0	3.4
U3 - Toxicity prior Beta(1, 1)	18.4	63.2	14.8	0.3	0	3.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.9	13.5	5.1	1.2	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	10.3	13.4	4.8	1.1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	9.3	13.6	5.5	1.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.7	13.7	5	1.2	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	8.2	14.1	6	1.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.6	14.2	5.6	1.3	0.1	
U2 - Toxicity prior Beta(1, 1)	8.3	13.9	6.1	1.4	0.1	
U3 - Toxicity prior Beta(1, 1)	8.5	14.2	5.7	1.3	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	5.3	1.6	0.6	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	5.1	1.5	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	5.2	1.6	0.6	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	5.1	1.5	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	4.7	1.7	0.6	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.6	1.6	0.6	0	
U2 - Toxicity prior Beta(1, 1)	0.8	4.6	1.7	0.6	0	
U3 - Toxicity prior Beta(1, 1)	0.8	4.6	1.6	0.6	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.6	52.6	28.4	7.5	0.2	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	13.6	55.6	24.2	5.8	0.2	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	10.2	49.5	32	7.1	0.4	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	12.8	53.9	26.6	5.9	0.2	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	49.8	31.4	8.8	0.6	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	12	53	26.9	7.4	0.2	0.5
U2 - Toxicity prior Beta(1, 1)	8.7	49.9	31.3	9	0.5	0.6
U3 - Toxicity prior Beta(1, 1)	12.8	52.4	26.8	7	0.3	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.4	11.6	8.4	3.7	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	5.8	11.9	8	3.5	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	5.2	11	8.6	4.1	1	
U3 - Toxicity prior Beta(0.3, 0.7)	5.5	11.4	8.2	3.9	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	10.7	8.7	4.6	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	5	11.2	8.4	4.3	1.1	
U2 - Toxicity prior Beta(1, 1)	4.7	10.7	8.7	4.6	1.2	
U3 - Toxicity prior Beta(1, 1)	5.1	11.2	8.3	4.3	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	4.2	2.4	1.4	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	4.2	2.3	1.4	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	4.1	2.5	1.5	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	4.1	2.4	1.4	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.8	2.4	1.6	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.8	2.4	1.5	0.4	
U2 - Toxicity prior Beta(1, 1)	0.4	3.8	2.4	1.5	0.4	
U3 - Toxicity prior Beta(1, 1)	0.5	3.8	2.4	1.4	0.4	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	52	0.5	0	0	0	47.5
U3 - Toxicity prior Beta(0.15, 0.85)	52.3	0.5	0	0	0	47.2
U2 - Toxicity prior Beta(0.3, 0.7)	52.8	0.5	0	0	0	46.7
U3 - Toxicity prior Beta(0.3, 0.7)	52.8	0.5	0	0	0	46.7
U2 - Toxicity prior Beta(0.5, 0.5)	53.5	0.3	0	0	0	46.2
U3 - Toxicity prior Beta(0.5, 0.5)	53.2	0.4	0	0	0	46.4
U2 - Toxicity prior Beta(1, 1)	53.8	0.4	0	0	0	45.8
U3 - Toxicity prior Beta(1, 1)	53.7	0.4	0	0	0	45.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.7	2.1	0	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	19.8	2.1	0	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	19.8	2.1	0	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	19.8	2.1	0	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	19.7	2.2	0	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	19.8	2.2	0	0	0	
U2 - Toxicity prior Beta(1, 1)	19.8	2.2	0	0	0	
U3 - Toxicity prior Beta(1, 1)	19.8	2.2	0	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U2 - Toxicity prior Beta(1, 1)	0.2	1.1	0	0	0	
U3 - Toxicity prior Beta(1, 1)	0.2	1.1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23.7	64.5	8.9	0.8	0	2.1
U3 - Toxicity prior Beta(0.15, 0.85)	26.2	61.5	9.2	0.9	0	2.2
U2 - Toxicity prior Beta(0.3, 0.7)	21.6	65.2	11.1	0.7	0	1.4
U3 - Toxicity prior Beta(0.3, 0.7)	25.1	61.5	11.2	0.7	0	1.5
U2 - Toxicity prior Beta(0.5, 0.5)	20.1	64	13.7	0.7	0	1.5
U3 - Toxicity prior Beta(0.5, 0.5)	23.9	60.8	13.3	0.6	0	1.4
U2 - Toxicity prior Beta(1, 1)	19.9	64	13.9	0.7	0	1.5
U3 - Toxicity prior Beta(1, 1)	25.4	60.1	12.2	0.7	0	1.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10	14	4.6	1.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	10.5	13.8	4.4	1.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	9.6	14	4.9	1.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	10.1	13.9	4.6	1.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	14.3	5.3	1.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	9.4	14.1	5	1.4	0.1	
U2 - Toxicity prior Beta(1, 1)	8.9	14.2	5.3	1.4	0	
U3 - Toxicity prior Beta(1, 1)	9.4	14.1	5	1.4	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	5.3	1.6	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	5.1	1.5	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	1	5.2	1.6	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	5.1	1.6	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1	4.9	1.7	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	4.8	1.6	0.5	0	
U2 - Toxicity prior Beta(1, 1)	1.1	4.9	1.7	0.5	0	
U3 - Toxicity prior Beta(1, 1)	1.2	4.8	1.6	0.5	0	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.7	2.2	0	0	0	81.1
U3 - Toxicity prior Beta(0.15, 0.85)	16.8	1.6	0	0	0	81.6
U2 - Toxicity prior Beta(0.3, 0.7)	17	2.2	0	0	0	80.8
U3 - Toxicity prior Beta(0.3, 0.7)	17.4	1.6	0	0	0	81
U2 - Toxicity prior Beta(0.5, 0.5)	16.9	1.9	0	0	0	81.2
U3 - Toxicity prior Beta(0.5, 0.5)	16.8	1.9	0	0	0	81.3
U2 - Toxicity prior Beta(1, 1)	17	1.7	0	0	0	81.3
U3 - Toxicity prior Beta(1, 1)	16.3	2.1	0	0	0	81.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.1	3.6	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	15.1	3.5	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	15.2	3.6	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	15.3	3.6	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	15.1	3.8	0.5	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	15.2	3.8	0.5	0	0	
U2 - Toxicity prior Beta(1, 1)	15.1	3.8	0.5	0	0	
U3 - Toxicity prior Beta(1, 1)	15.1	3.8	0.5	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.2	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.2	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	2.2	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2.2	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	2.2	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	2.1	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.6	2.2	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.6	2.1	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	0.9	2.3	53.9	42.1	0
U3 - Toxicity prior Beta(0.15, 0.85)	1.3	1.3	3.6	58.7	35.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	0.8	2.3	53.6	42.6	0
U3 - Toxicity prior Beta(0.3, 0.7)	1.3	1.4	3.4	57.5	36.4	0
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	0.9	2.8	53.1	42.5	0
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	1.5	3.6	57.1	36.6	0
U2 - Toxicity prior Beta(1, 1)	0.9	0.9	3.1	53.4	41.7	0
U3 - Toxicity prior Beta(1, 1)	1.4	1.5	3.7	57.3	36.1	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	3.8	12.9	7	
U3 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	3.9	13.4	6.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	3.8	12.9	7.1	
U3 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	3.9	13.4	6.4	
U2 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	3.8	12.9	7.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	3.2	3.9	13.4	6.5	
U2 - Toxicity prior Beta(1, 1)	3.1	3.2	3.8	12.9	7.1	
U3 - Toxicity prior Beta(1, 1)	3.1	3.2	3.9	13.4	6.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1	2.4	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.4	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.4	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1	2.4	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.4	1.4	
U2 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.4	1.5	
U3 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.4	1.4	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	28.5	40.9	29	1.5	0	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	32.7	39.2	26.2	1.7	0	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	27	40.2	30.2	2.4	0.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	31.9	38.6	27.6	1.8	0	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	24.8	36.6	36.2	2.2	0.1	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	29.8	36.5	31.4	2.2	0	0.1
U2 - Toxicity prior Beta(1, 1)	25	37.2	35.3	2.3	0.1	0.1
U3 - Toxicity prior Beta(1, 1)	30.5	36.1	30.8	2.5	0	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.1	11.2	8	1.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	11.2	7.6	1.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	11	8.3	1.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.4	11.1	7.8	1.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	8.1	11	8.9	1.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.7	11.1	8.5	1.7	0.1	
U2 - Toxicity prior Beta(1, 1)	8	11	9	1.9	0.1	
U3 - Toxicity prior Beta(1, 1)	8.6	11.1	8.5	1.7	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.1	2.1	0.7	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.1	2	0.7	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.1	2.2	0.8	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.1	2.1	0.7	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	3.8	2.2	0.8	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.8	2.1	0.7	0	
U2 - Toxicity prior Beta(1, 1)	0.8	3.8	2.2	0.8	0	
U3 - Toxicity prior Beta(1, 1)	0.9	3.8	2.1	0.7	0	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	27.5	43.4	21.8	1.9	0	5.4
U3 - Toxicity prior Beta(0.15, 0.85)	31.8	41.3	19.2	2.1	0	5.6
U2 - Toxicity prior Beta(0.3, 0.7)	25.4	42.8	24.9	2.5	0	4.4
U3 - Toxicity prior Beta(0.3, 0.7)	30.4	43.1	19.5	2.4	0	4.6
U2 - Toxicity prior Beta(0.5, 0.5)	22.7	42.9	28.3	2.7	0	3.4
U3 - Toxicity prior Beta(0.5, 0.5)	25.9	42.8	24.8	2.7	0	3.8
U2 - Toxicity prior Beta(1, 1)	23.1	42.8	27.5	3	0	3.6
U3 - Toxicity prior Beta(1, 1)	25.7	43.9	23.6	3.1	0	3.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.7	10.4	6.5	1.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	11.4	10.2	6.2	1.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	10.3	10.4	6.9	2	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	11.1	10.2	6.4	1.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	9	10.6	7.7	2.2	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	9.6	10.6	7.3	2.1	0.1	
U2 - Toxicity prior Beta(1, 1)	9	10.6	7.8	2.2	0.1	
U3 - Toxicity prior Beta(1, 1)	9.5	10.6	7.4	2.1	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.6	1.6	0.9	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.5	1.5	0.8	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.5	1.7	0.9	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.4	1.5	0.9	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	4.1	1.8	1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.1	1.7	0.9	0	
U2 - Toxicity prior Beta(1, 1)	0.8	4.1	1.8	1	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	4.1	1.7	0.9	0	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.5	34.7	41.3	5.1	0	1.4
U3 - Toxicity prior Beta(0.15, 0.85)	21.3	35.4	37.6	4.3	0	1.4
U2 - Toxicity prior Beta(0.3, 0.7)	15.4	33.4	44.2	5.9	0	1.1
U3 - Toxicity prior Beta(0.3, 0.7)	20	33.7	39.9	5.2	0	1.2
U2 - Toxicity prior Beta(0.5, 0.5)	14.3	30.5	46	8.4	0	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	17.4	32.8	42.1	7	0	0.7
U2 - Toxicity prior Beta(1, 1)	14.1	31.3	46.6	7.3	0	0.7
U3 - Toxicity prior Beta(1, 1)	18.7	32	41.9	6.7	0	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.8	9.8	9.3	2.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	8.3	9.8	9	2.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	7.4	9.5	9.7	3	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8	9.5	9.4	2.8	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	6.5	9.4	10.3	3.5	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	7	9.4	10	3.3	0.2	
U2 - Toxicity prior Beta(1, 1)	6.4	9.4	10.4	3.4	0.3	
U3 - Toxicity prior Beta(1, 1)	7	9.4	10.1	3.2	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.2	2.1	1.1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4.2	2.1	1.1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	4.1	2.2	1.2	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	4.1	2.1	1.1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.8	2.2	1.2	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.7	2.2	1.2	0.1	
U2 - Toxicity prior Beta(1, 1)	0.6	3.7	2.3	1.2	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.7	2.2	1.1	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.4	37.2	2.8	0	0	30.6
U3 - Toxicity prior Beta(0.15, 0.85)	30.7	35.6	2.7	0	0	31
U2 - Toxicity prior Beta(0.3, 0.7)	27.1	40.6	3.1	0	0	29.2
U3 - Toxicity prior Beta(0.3, 0.7)	28.7	39.3	3.5	0	0	28.5
U2 - Toxicity prior Beta(0.5, 0.5)	26.3	44.8	4.2	0.1	0	24.6
U3 - Toxicity prior Beta(0.5, 0.5)	28.3	43.5	3.6	0	0	24.6
U2 - Toxicity prior Beta(1, 1)	25.8	44.6	4.4	0.1	0	25.1
U3 - Toxicity prior Beta(1, 1)	28.7	42.7	3.6	0	0	25
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	14.1	10.4	2.4	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	14.4	10.3	2.2	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	13.6	10.8	2.5	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	13.9	10.8	2.4	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	12.4	11.8	3.2	0.5	0	
U3 - Toxicity prior Beta(0.5, 0.5)	12.8	11.7	3	0.5	0	
U2 - Toxicity prior Beta(1, 1)	12.5	11.7	3.2	0.5	0	
U3 - Toxicity prior Beta(1, 1)	12.8	11.6	3	0.5	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	4.8	0.8	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.7	0.8	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	4.8	0.9	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.8	0.9	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	1	4.4	1.1	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	4.3	0.9	0.2	0	
U2 - Toxicity prior Beta(1, 1)	1	4.4	1	0.2	0	
U3 - Toxicity prior Beta(1, 1)	1	4.3	1	0.2	0	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	60.7	34.7	2	0	0	2.6
U3 - Toxicity prior Beta(0.15, 0.85)	64.4	31.9	1	0	0	2.7
U2 - Toxicity prior Beta(0.3, 0.7)	59.1	36.2	2.1	0	0	2.6
U3 - Toxicity prior Beta(0.3, 0.7)	62.9	33.4	1.1	0	0	2.6
U2 - Toxicity prior Beta(0.5, 0.5)	54.4	39.5	3.6	0	0	2.5
U3 - Toxicity prior Beta(0.5, 0.5)	59.9	35.5	2.2	0	0	2.4
U2 - Toxicity prior Beta(1, 1)	55.6	38.8	3.1	0	0	2.5
U3 - Toxicity prior Beta(1, 1)	59.6	35.8	2.1	0	0	2.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.3	9.7	1.3	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	19	9.5	0.9	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	18.2	9.9	1.3	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	18.8	9.6	0.9	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17.2	10.6	1.6	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	18	10.4	1.1	0.1	0	
U2 - Toxicity prior Beta(1, 1)	17.2	10.6	1.5	0.1	0	
U3 - Toxicity prior Beta(1, 1)	18	10.3	1.1	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	3.6	0.7	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	1	3.4	0.4	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	3.6	0.7	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	1	3.5	0.4	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	3.4	0.8	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.3	0.5	0	0	
U2 - Toxicity prior Beta(1, 1)	0.9	3.5	0.7	0.1	0	
U3 - Toxicity prior Beta(1, 1)	1	3.3	0.5	0	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15	17	18.5	39.7	9.8	0
U3 - Toxicity prior Beta(0.15, 0.85)	20.1	19.6	17.1	35.9	7.3	0
U2 - Toxicity prior Beta(0.3, 0.7)	14.2	16.7	16.9	41.4	10.8	0
U3 - Toxicity prior Beta(0.3, 0.7)	19.6	20.7	14.6	36.9	8.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	13.6	16.2	13.8	44.3	12.1	0
U3 - Toxicity prior Beta(0.5, 0.5)	19.9	19.4	11.7	39.1	9.9	0
U2 - Toxicity prior Beta(1, 1)	13.6	15.5	13.9	44.1	12.9	0
U3 - Toxicity prior Beta(1, 1)	20.1	18.1	13.8	38.1	9.9	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.7	6.1	6.8	9.5	3	
U3 - Toxicity prior Beta(0.15, 0.85)	5	6.3	6.9	9.4	2.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.5	5.8	6.6	9.8	3.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	6.1	6.6	9.8	2.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	5.6	6.4	10.3	3.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	5.8	6.4	10.3	3.1	
U2 - Toxicity prior Beta(1, 1)	4.2	5.6	6.4	10.3	3.6	
U3 - Toxicity prior Beta(1, 1)	4.4	5.8	6.4	10.3	3.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3	2	2.3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.1	2.1	2.2	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3	2	2.3	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3	2	2.2	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	2	2.2	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	2	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.9	2	2.2	1.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.9	2	2.2	1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	2.6	43.8	34	18	1.6	0
U3 - Toxicity prior Beta(0.15, 0.85)	3.1	48.9	31.9	15	1.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	2.4	43.6	34	18	2	0
U3 - Toxicity prior Beta(0.3, 0.7)	3.3	48.4	32.9	14.3	1.1	0
U2 - Toxicity prior Beta(0.5, 0.5)	2.3	43.6	34.5	17.4	2.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	3.2	48.2	33.1	14.4	1.1	0
U2 - Toxicity prior Beta(1, 1)	2.4	43.3	34.1	17.8	2.4	0
U3 - Toxicity prior Beta(1, 1)	3.3	47.9	33.3	14.1	1.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4	14.5	8.2	3.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	15.3	7.7	2.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4	14.3	8.2	3.1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	15.2	7.7	2.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	3.9	14.3	8.3	3.1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4	15.3	7.7	2.8	0.3	
U2 - Toxicity prior Beta(1, 1)	3.9	14.3	8.4	3.1	0.4	
U3 - Toxicity prior Beta(1, 1)	4	15.2	7.8	2.7	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.9	1.8	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.9	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.9	1.8	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.9	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.9	1.8	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.8	1.7	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.9	1.8	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.8	1.7	0.7	0.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.8	14.3	33	33.9	10	0
U3 - Toxicity prior Beta(0.15, 0.85)	14	18.5	31.2	30.6	5.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	10.2	12.3	34	33.4	10.1	0
U3 - Toxicity prior Beta(0.3, 0.7)	14.5	18.1	31.7	29.5	6.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	12.7	32	34.6	11.9	0
U3 - Toxicity prior Beta(0.5, 0.5)	14.4	17.6	32.5	29	6.5	0
U2 - Toxicity prior Beta(1, 1)	9.9	12.4	31.7	34.8	11.2	0
U3 - Toxicity prior Beta(1, 1)	14.5	17.4	31.4	30.4	6.3	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	4.6	8.7	9.1	4.1	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	5	9	9	3.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	4.4	8.5	9.2	4.4	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	4.8	8.9	9.1	3.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	4.3	8.2	9.4	4.6	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	4.7	8.7	9.3	3.8	
U2 - Toxicity prior Beta(1, 1)	3.4	4.4	8.3	9.4	4.6	
U3 - Toxicity prior Beta(1, 1)	3.4	4.6	8.7	9.4	3.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	2	2.9	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.9	2.1	2.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	2	2.9	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	2.1	2.8	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	2	2.8	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.8	2.1	2.8	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.7	2	2.8	1.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.8	2.1	2.8	1.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.1	8	14.6	48.1	22.2	0
U3 - Toxicity prior Beta(0.15, 0.85)	11.4	8.8	13.2	48.7	17.9	0
U2 - Toxicity prior Beta(0.3, 0.7)	6.6	6	15.1	48.1	24.2	0
U3 - Toxicity prior Beta(0.3, 0.7)	11.3	7.2	14	46.7	20.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	6.2	6.3	12.3	48.1	27.1	0
U3 - Toxicity prior Beta(0.5, 0.5)	9.9	8.2	12.3	46.8	22.8	0
U2 - Toxicity prior Beta(1, 1)	6.1	6	11.9	47.7	28.3	0
U3 - Toxicity prior Beta(1, 1)	9.8	8.4	12.9	45.8	23.1	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.9	6.4	11.1	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	3.4	4	6.5	11.2	4.9	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.8	5.9	11.3	5.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.4	3.9	6.1	11.4	5.2	
U2 - Toxicity prior Beta(0.5, 0.5)	3.2	3.7	5.6	11.6	5.9	
U3 - Toxicity prior Beta(0.5, 0.5)	3.3	3.8	5.7	11.8	5.4	
U2 - Toxicity prior Beta(1, 1)	3.3	3.7	5.5	11.5	6	
U3 - Toxicity prior Beta(1, 1)	3.3	3.8	5.8	11.7	5.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.9	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.8	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.7	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.7	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.6	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.6	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.6	1.3	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.5	2.6	1.2	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.9	22.2	46.2	14.2	2.2	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	19.9	21.7	43.5	13.1	1.4	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	13.5	19.6	47.6	16.4	2.6	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	18.9	22.1	43.8	13.1	1.8	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	12.6	17.8	48.1	18.6	2.7	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	18.3	19.9	44.4	15	2.1	0.3
U2 - Toxicity prior Beta(1, 1)	12.6	17.9	47.5	18.9	2.8	0.3
U3 - Toxicity prior Beta(1, 1)	17.9	20.4	44.7	14.7	2	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.8	7.6	10	5.2	1.3	
U3 - Toxicity prior Beta(0.15, 0.85)	6.4	7.6	9.8	5.1	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	5.5	7.2	10.2	5.6	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	6.1	7.2	10	5.5	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.1	6.9	10.1	6.3	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	5.5	6.9	9.9	6.1	1.4	
U2 - Toxicity prior Beta(1, 1)	5.1	6.9	10.1	6.3	1.6	
U3 - Toxicity prior Beta(1, 1)	5.6	7	9.9	6.1	1.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.8	2.3	1.8	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.7	2.2	1.7	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	2.3	1.8	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.7	2.3	1.8	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	2.3	1.8	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.5	2.2	1.7	0.5	
U2 - Toxicity prior Beta(1, 1)	0.5	3.4	2.2	1.8	0.5	
U3 - Toxicity prior Beta(1, 1)	0.7	3.5	2.2	1.8	0.5	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	6.7	47.1	37.9	7.5	0.8	0
U3 - Toxicity prior Beta(0.15, 0.85)	9.5	52.1	32.2	5.8	0.4	0
U2 - Toxicity prior Beta(0.3, 0.7)	6.6	44.8	38.2	9.4	1	0
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	50.6	32.3	7.2	0.3	0
U2 - Toxicity prior Beta(0.5, 0.5)	5.9	41.9	40.9	10.6	0.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	9.3	48.6	33.4	8.3	0.3	0.1
U2 - Toxicity prior Beta(1, 1)	6.9	43.1	39.3	10.1	0.6	0
U3 - Toxicity prior Beta(1, 1)	9.8	47.8	33.6	8.3	0.4	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.5	13.9	9.1	2.3	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	4.7	14.9	8.2	2.1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	13.7	9.1	2.5	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	14.7	8.1	2.3	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	13.4	9.1	2.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	14.6	8.2	2.5	0.2	
U2 - Toxicity prior Beta(1, 1)	4.2	13.6	9.1	2.8	0.3	
U3 - Toxicity prior Beta(1, 1)	4.4	14.5	8.3	2.5	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.3	2.3	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	2	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	2.3	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	2	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3.1	2.3	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.1	2.1	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.2	3.1	2.3	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.1	2.1	0.7	0.1	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.4	46.8	11.7	2.4	0.6	12.1
U3 - Toxicity prior Beta(0.15, 0.85)	29	46.9	8.4	2.3	0.4	13
U2 - Toxicity prior Beta(0.3, 0.7)	23.7	48.5	13.4	3	0.6	10.8
U3 - Toxicity prior Beta(0.3, 0.7)	26.7	49.3	10.3	2.3	0.5	10.9
U2 - Toxicity prior Beta(0.5, 0.5)	21.9	51	14.9	3.4	0.5	8.3
U3 - Toxicity prior Beta(0.5, 0.5)	25.4	51	11.3	3	0.3	9
U2 - Toxicity prior Beta(1, 1)	21.5	50.3	14.9	3.9	0.5	8.9
U3 - Toxicity prior Beta(1, 1)	25	51.3	11.6	2.4	0.4	9.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.1	11.1	4.6	2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	11.6	11	4.3	1.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	10.5	11.1	4.9	2.3	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	11.1	11.1	4.6	2.1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	9.5	11.2	5.6	2.6	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	10.1	11.2	5.2	2.5	0.5	
U2 - Toxicity prior Beta(1, 1)	9.6	11.1	5.6	2.6	0.5	
U3 - Toxicity prior Beta(1, 1)	10	11.2	5.2	2.5	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	5.2	1.4	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	5.1	1.3	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	5.2	1.4	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	5.1	1.4	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	4.7	1.6	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	1	4.7	1.4	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.9	4.7	1.5	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	1	4.7	1.4	0.9	0.2	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.8	57.7	11	0.8	0	4.7
U3 - Toxicity prior Beta(0.15, 0.85)	30.5	56	7.6	0.7	0	5.2
U2 - Toxicity prior Beta(0.3, 0.7)	25.2	57.5	12.5	0.7	0	4.1
U3 - Toxicity prior Beta(0.3, 0.7)	28.1	57.8	8.9	0.9	0	4.3
U2 - Toxicity prior Beta(0.5, 0.5)	21.8	60.9	12.9	1	0	3.4
U3 - Toxicity prior Beta(0.5, 0.5)	26.6	58	11	1.2	0	3.2
U2 - Toxicity prior Beta(1, 1)	22.3	60.8	12.5	0.7	0	3.7
U3 - Toxicity prior Beta(1, 1)	27.2	58.3	10.1	0.8	0	3.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.3	12.6	4.8	1.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	10.9	12.4	4.5	1.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	9.8	12.5	5.1	2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	10.5	12.3	4.8	1.9	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	12.8	5.5	2.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	9.6	12.4	5.3	2.1	0.4	
U2 - Toxicity prior Beta(1, 1)	8.9	12.7	5.6	2.2	0.4	
U3 - Toxicity prior Beta(1, 1)	9.5	12.5	5.3	2.1	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	5.6	1.5	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	5.5	1.5	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1	5.5	1.6	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1	5.4	1.5	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1	5.1	1.6	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	5	1.5	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	1	5.1	1.6	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	1.1	5.1	1.5	0.8	0.1	